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HUJI Applicative Research Report — Week 46, 2026

4 high-potential paper(s) · Weekly · Week 46, 2026 · Generated November 09, 2026

This week's digest highlights groundbreaking research from Hebrew University poised for significant commercial impact across diverse sectors. We feature innovations in microbiome-based therapeutics targeting metabolic diseases, alongside a novel pump-free microfluidic platform enhancing drug discovery. Also showcased is a breakthrough in controlling nanocrystal chirality for advanced materials, and a crucial discovery in bacterial pathogenesis opening new avenues for anti-infective treatments. These advancements present timely opportunities for investors and industry partners seeking to revolutionize health, materials science, and pharmaceutical development.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Kfir Sharabi** — Drug Discovery, Clinical, FoodTech
- **Ofra Benny** — Drug Discovery, Medical Device, Diagnostics
- **David Avnir** — Materials, Drug Discovery, Other
- **Emanuel Hanski** — Drug Discovery, Immunology, Proteomics, Clinical

ALL BRANCHES — RESEARCH HIGHLIGHTS

1. Microbiota-derived indole-3-propionic acid regulates glucose homeostasis via remodeling of hepatic mitochondrial metabolism. 31/50

"Gut metabolite discovery offers new precision targets for diabetes and metabolic syndrome treatments."

Main researcher: Kfir Sharabi

The Institute of Biochemistry, Food Science, and Nutrition, The Robert H. Smith Faculty of Agriculture, The Hebrew University of Jerusalem, Rehovot, Israel.

Published in Gut microbes · 2026 Nov 6

ABOUT THIS RESEARCH

Researchers have identified that a gut microbiota metabolite called indole-3-propionic acid (IPA) improves blood sugar regulation by directly altering how liver mitochondria produce energy. This discovery reveals a specific signaling pathway through which the gut microbiome influences systemic metabolic health and diabetes progression.

COMMERCIAL ANGLE

This mechanism offers a new target for the development of microbiome-based therapeutics or precision nutritional supplements aimed at treating type 2 diabetes and metabolic syndrome.

COMMERCIALISATION METRICS

Novelty 7/10

The research provides a high-quality mechanistic link between a specific microbial metabolite and metabolic disease, representing a significant advancement in the field of microbiome-gut-liver axis therapeutics. While it builds upon existing knowledge of indole-derivatives, its potential for targeted metabolic interventions is highly relevant for drug development, though it does not yet constitute a broad, transformative technological platform.

Commercial Potential 8/10

Indole-3-propionic acid is a well-characterized microbial metabolite with a clear translational pathway into metabolic disease therapeutics or precision-nutrition supplements. The mechanistic focus on hepatic mitochondrial remodeling provides a specific, druggable therapeutic target and a high-potential platform for developing novel therapies for type 2 diabetes and metabolic syndrome.

Market Size 9/10

Metabolic disorders like Type 2 diabetes and NAFLD represent a massive, multi-billion dollar global market, and a microbiota-derived metabolite offers a high-value, platform-level opportunity for novel pharmacological or nutritional interventions.

Tech Readiness 3/10

This study identifies a molecular mechanism for glucose regulation in a pre-clinical model, representing basic research that is far from clinical translation or commercial development. Significant work remains in establishing causality, toxicology, and human efficacy before it can be considered a viable therapeutic candidate.

IP Strength 4/10

The paper describes a biological mechanism involving a naturally occurring metabolite (IPA), which is generally not patentable. Defensibility would rely on specific downstream drug formulations or synthetic analogs, which are not the focus of this mechanistic study.

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2. A pump-free microfluidic device for integrated multi-functional testing of tumor spheroids. 30/50

"Novel pump-free organ-on-a-chip simplifies drug testing, improving accuracy and reducing R&D costs."

Main researcher: Ofra Benny

The Institute for Drug Research, The School of Pharmacy, Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel.

Published in APL bioengineering · 2026-06

ABOUT THIS RESEARCH

Researchers have developed a microfluidic platform that uses passive mechanisms instead of external pumps to study 3D tumor spheroids. This device allows for more integrated and simplified testing of how cancer models react to different environments or drugs.

COMMERCIAL ANGLE

The technology can be commercialized as a low-cost, plug-and-play organ-on-a-chip platform for pharmaceutical companies to improve the accuracy of drug toxicity and efficacy screening.

COMMERCIALISATION METRICS

Novelty 6/10

While the elimination of external pumps simplifies microfluidic architecture, the integration of tumor spheroid testing via passive fluidics is an incremental engineering improvement rather than a paradigm-shifting scientific discovery. The novelty lies in the device's specific configuration for ease of use, but it builds upon well-established principles of capillary-driven or gravity-driven microfluidics.

Commercial Potential 7/10

The device addresses a significant bottleneck in drug discovery (high-throughput 3D modeling) and the 'pump-free' aspect suggests a scalable, cost-effective hardware platform. However, commercial success depends on navigating the highly competitive organ-on-a-chip market and demonstrating clear workflow integration for pharmaceutical R&D.

Market Size 7/10

The technology targets the high-growth organ-on-a-chip and drug discovery market, offering significant value by reducing infrastructure costs and improving high-throughput screening scalability. However, it is a specialized tool within a niche segment of the broader preclinical testing market rather than a broad-scale clinical diagnostic.

Tech Readiness 4/10

The paper describes the fundamental development and proof-of-concept of a microfluidic prototype, which is currently at the laboratory validation stage. While it addresses a significant need in drug testing, it lacks the standardized manufacturing and rigorous biological validation required for near-market or industrial adoption.

IP Strength 6/10

The innovation lies in the physical architecture of a passive microfluidic system, which is patentable, but the lack of a proprietary reagent or biological readout limits its platform defensibility against broader microfluidic competitors.

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3. Prochiral nanocrystals. 28/50

"Breakthrough in nanocrystal chirality control promises advanced materials for drugs, sensors, and semiconductors."

Main researcher: David Avnir · david.avnir@mail.huji.ac.il.

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Published in Nanoscale · 2026 Nov 6

ABOUT THIS RESEARCH

The research focuses on the creation and manipulation of nanocrystals that lack inherent chirality but can be induced to exhibit chiral properties. This work likely explores how symmetry breaking at the nanoscale can be controlled for specific physical or chemical functions.

COMMERCIAL ANGLE

Controlling nanocrystal chirality offers significant potential for developing advanced chiral materials for enantiopure drug manufacturing, specialized optical sensors, and next-generation semiconductor components.

COMMERCIALISATION METRICS

Novelty 9/10

The development of prochiral nanocrystals represents a fundamental paradigm shift in materials science, enabling site-specific chemical reactivity and enantioselective catalysis on a nanometric scale. This platform-enabling technology creates a novel category of chiral building blocks with significant commercial potential in asymmetric synthesis and pharmaceutical manufacturing.

Commercial Potential 4/10

While prochiral nanocrystals have interesting applications in asymmetric catalysis and sensing, the lack of specific functional data or a defined problem statement makes it difficult to assess the scalability or market readiness of the technology. It remains a fundamental materials science advancement that requires significant downstream optimization to reach industrial or clinical utility.

Market Size 6/10

While prochirality is a foundational challenge in asymmetric catalysis and pharmaceutical manufacturing, the technology is currently at a fundamental material science stage with long-term, unquantified commercial pathing. The potential market value is high due to the pervasiveness of chirality in drug development, but the lack of demonstrated industrial scalability limits its current market-size score.

Tech Readiness 2/10

The study of prochiral nanocrystals is fundamental materials science focused on structural symmetry and surface chemistry, placing it firmly at the early benchtop proof-of-concept phase. There is no clear line of sight to a specific industrial or clinical application at this stage of research.

IP Strength 7/10

Prochiral nanocrystals represent a highly specialized materials science platform with significant potential for proprietary control over synthesis and surface modification, though actual defensibility is contingent upon the breadth of the specific composition-of-matter and application-specific patents.

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4. Group A Streptococcus host-pathogen dual crosstalk. 28/50

"Disrupting bacterial metabolic hijacking offers a new small-molecule strategy against Group A Strep infections."

Main researcher: Emanuel Hanski · emanuelh@ekmd.huji.ac.il.

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Published in Trends in microbiology · 2026-06

ABOUT THIS RESEARCH

Researchers have discovered that Group A Streptococcus bacteria hijack the human cell's stress response to force the production of the amino acid asparagine, which the bacteria then consume to fuel infection. This mechanism creates a metabolic bridge between host cell stress and bacterial virulence, mimicking similar pathways seen in cancer cells.

COMMERCIAL ANGLE

The discovery of the PERK-eIF2α-ATF4-Asn axis provides a novel metabolic target for developing small-molecule inhibitors to starve bacterial pathogens or disrupt their ability to exploit host resources.

COMMERCIALISATION METRICS

Novelty 8/10

The research identifies a highly specific, previously unknown metabolic hijacking mechanism where an extracellular bacterium exploits a host stress pathway (PERK-eIF2α-ATF4) to drive nutrient biosynthesis. This discovery of a direct 'crosstalk' circuit provides a novel, targetable metabolic vulnerability that parallels oncogenic pathways, offering significant potential for new antimicrobial strategies.

Commercial Potential 6/10

The identification of a specific metabolic axis (ATF4-Asn) provides a clear target for novel antimicrobial drug development, though the transition from mechanistic discovery to a viable therapeutic lead remains speculative.

Market Size 7/10

The research targets Group A Streptococcus, a major global pathogen with high clinical burden, but its most significant market value lies in the potential to translate the ATF4-Asn metabolic axis into oncology therapeutics. While the infectious disease application is substantial, the broader addressable market is expanded by the cross-sector relevance to cancer metabolism and therapy resistance.

Tech Readiness 2/10

The paper describes fundamental mechanistic biology and discovery of a signaling pathway in a laboratory setting. It lacks any mention of therapeutic intervention, drug development, or preclinical validation required to move beyond basic research.

IP Strength 5/10

The paper describes a fundamental biological mechanism (the ATF4-Asn metabolic axis) rather than a specific therapeutic molecule or engineered platform. While the discovery provides a targetable pathway, the core finding is a natural phenomenon which is difficult to patent without specific, non-obvious chemical or biological interventions.

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